

The University of Hong Kong
School of Computing and Data Science

DASC7011 Statistical Inference for Data Science

Course Description

Computing power has revolutionized the theory and practice of statistical inference. Reciprocally, novel statistical inference procedures are becoming an integral part of data science. By focusing on the interplay between statistical inference and methodologies for data science, this course reviews the main concepts underpinning classical statistical inference, studies computer-intensive methods for conducting statistical inference, and examines important issues concerning statistical inference drawn upon modern learning technologies. Contents include method of moments estimation (MME), least squares estimation (LSE), maximum likelihood estimation (MLE), principal component analysis (PCA), singular value decomposition (SVD), simple Monte Carlo (MC) simulation, jackknife and bootstrap resampling methods, and LASSO and RIDGE regressions.

Course Objectives

The course objectives are:

1. To gain an understanding of classical statistical inference including the method-of-moments, least squares, and likelihood methodologies.
2. To gain an understanding of early age computer aided methodologies for statistical inference including the PCA, SVD, MC, jackknife and bootstrap algorithms.
3. To gain an understanding of several selected topics on modern methodologies for statistical inference including the ridge and lasso regression.

Course Learning Outcomes

Upon completion, students should be able to

1. Understand and conduct classical frequentism inference including estimating and hypothesis testing.
2. Understand and conduct Method-of-moment estimation.
3. Understand and conduct Least Squares estimation.
4. Understand and conduct likelihood inference.
5. Understand and conduct PCA and SVD.
6. Understand and conduct simple Monte Carlo simulation.

7. Conduct nonparametric bootstrap, and parametric bootstrap for some simple cases.
8. Understand and conduct techniques on modelling high-dimensional data using the ridge and lasso regressions.

Assessment

Assessment includes one examination (60% weighting) and a course work assessment (40% weighting) based on 3 assignments and 1 class test. Any assignment submission passing the due date will be penalized. Partially or wholly copied assignments will be penalized and/or reported as plagiarism.

Textbook

Bradley Efron & Trevor Hastie: *Computer Age Statistical Inference: Algorithms, Evidence, and Data Science*, Cambridge, 2016.

References

- [1.] Krzysztof Podgórski: *Lecture Notes on Statistical Inference*, University of Limerick, Ireland, 2009.
- [2.] Suhasini Subba Rao: *Advanced Statistical Inference* (E-Copy), Texas A&M University, U.S., 2017.
- [3.] Dirk P. Kroese, Zdravko I. Botev, Thomas Taimre, Radislav Vaisman: *Data Science and Machine Learning: Mathematical and Statical Methods* (E-Copy), Chapman & Hall/CRC, 2020.

Department's policy on absence from class test

If for any reason you are or have been unable to attend a mid-term/class test, and if you wish to have a supplementary mid-term/class test,

(a) all full-time students (including MStat students) should write to the General Office of the Department of Statistics and Actuarial Science giving reasons for your absence;

(b) all part-time students should write to the course instructor giving reasons for your absence, within 7 days of the absence.

A special/supplementary test is normally granted to those absent from the original test due to illness and with original medical certificate provided. Students absent due to other reasons are not granted a special/supplementary test unless with very special circumstances and with valid documental proofs provided.

University's Policy on Student Plagiarism in Undergraduate and Taught Postgraduate Curricula

Please refer to the University's website: <https://tl.hku.hk/plagiarism/>